

AWS

NETWORKING

FUNDAMENTALS

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REGIONS & AVAILABILITY ZONES

REGION

A Region is a geographic location where AWS operates a set of isolated infrastructure.

Ex. `ap-south-1` → Mumbai (India)

- Regions are physically separated from each other
- Each Region operates independently
- Designed to limit the blast radius of failures
- You typically deploy applications in one Region

AVAILABILITY ZONE

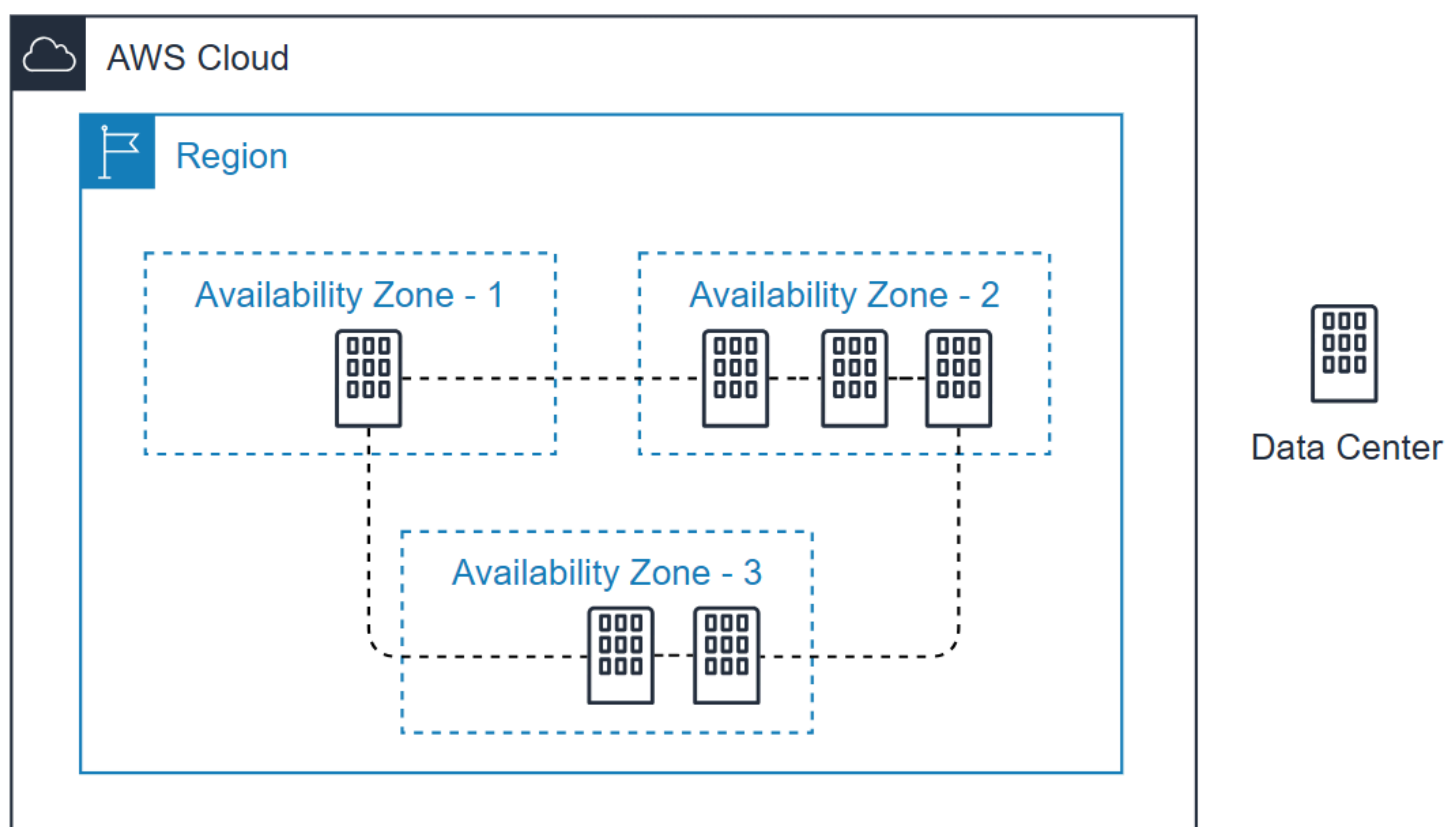
An Availability Zone is one or more physically separate data centers within a Region.

Ex. `ap-south-1a`

`ap-south-1b`

`ap-south-1c` (AZs in `ap-south-1` Region)

- Each AZ has independent power, cooling, and networking
- AZs are isolated from failures in other AZs
- AZs are connected using low-latency, high-bandwidth private links
- High availability in AWS is achieved by running applications across multiple AZs (e.g., `ap-south-1a`, `ap-south-1b`) to tolerate single-AZ failures.



VPCS & SUBNETS

VPC (VIRTUAL PRIVATE CLOUD)

A VPC is a logically isolated virtual network in AWS where you run your cloud resources.

- A VPC is created inside a single Region
- You define an IP address range (CIDR block) for the VPC
- A VPC spans all Availability Zones in the Region
- Resources inside a VPC communicate using private IP addresses
- You control routing, security, and connectivity

SUBNET

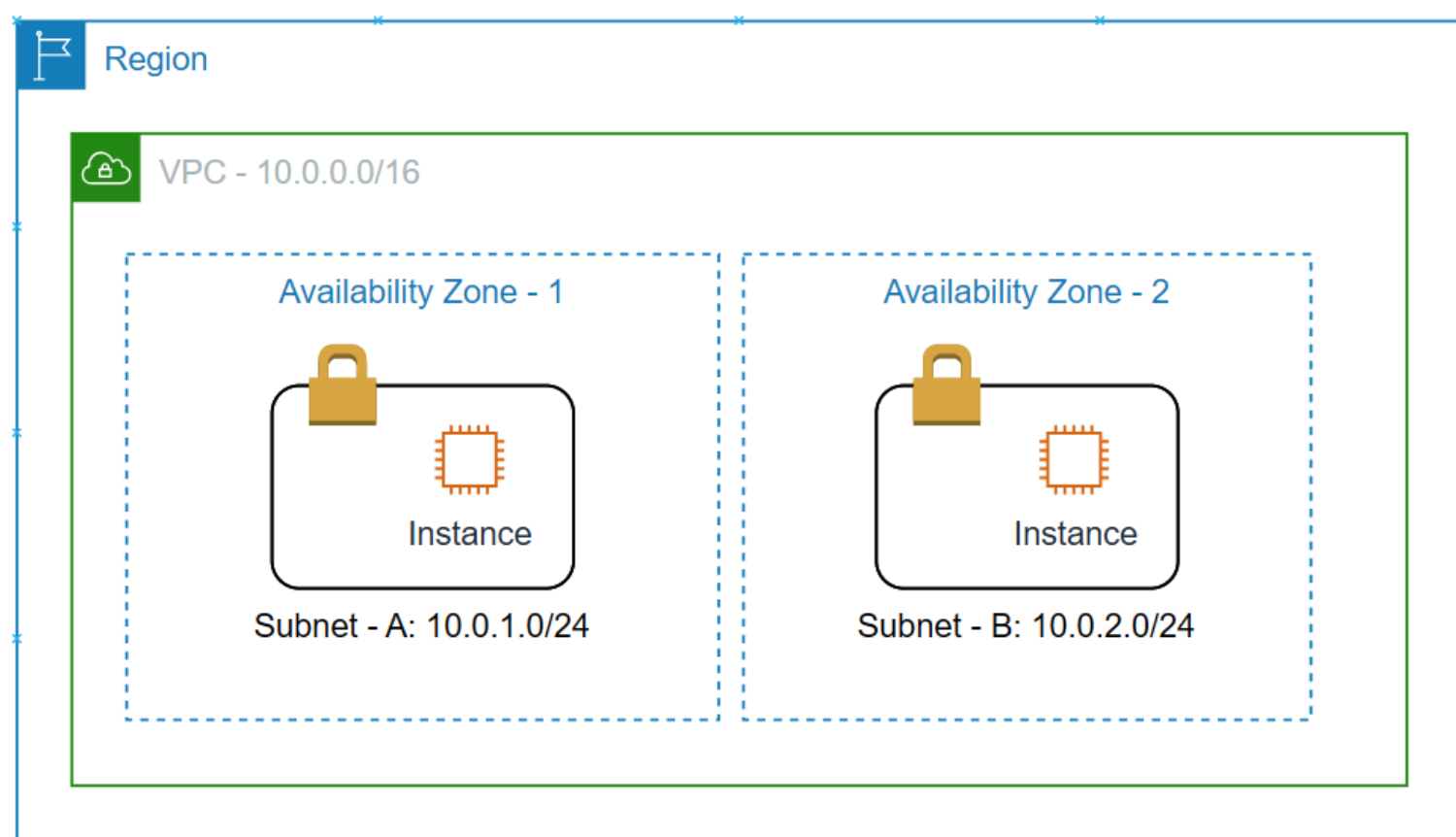
A subnet is a range of IP addresses within a VPC that exists in exactly one Availability Zone.

- A subnet is created inside a VPC
- Each subnet belongs to one and only one AZ
- Subnets are defined using CIDR blocks
- Subnet CIDR ranges must not overlap
- AWS resources (EC2, RDS, ALB targets) are placed inside subnets

TYPES OF SUBNET

Public Subnet: A subnet whose route table includes a route to an Internet Gateway (IGW), allowing resources to be reachable from the internet.

Private Subnet: A subnet without a direct route to an Internet Gateway, preventing direct internet access to its resources.

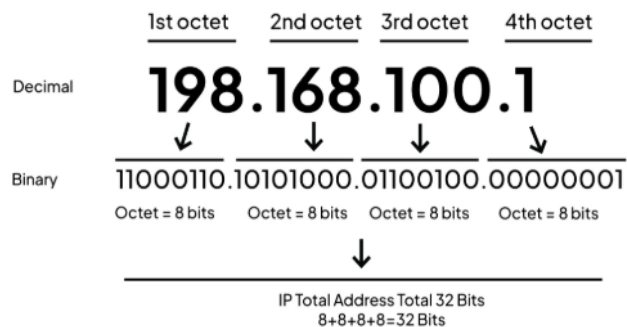


IPv4 & CIDR

IPV4 ADDRESS

A 32-bit numeric address used to identify a device or resource on a network.

Written as **x.x.x.x**, where each x is an octet (8 bits, 0–255).



CIDR BLOCK (CLASSLESS INTER-DOMAIN ROUTING)

A CIDR block defines a range of IP addresses.

Written as: **x.x.x.x/n**

n → number of fixed **network bits** (the “frozen” part)

32 - n → number of **variable host bits** (assignable to devices)

CIDR Example: 10.0.1.0/24

/24 → first **24 bits** are network, last **8 bits** are host

Network bits(frozen) | Host bits(can be assigned to device)

11111111 . 11111111 . 11111111 . 00000000



- Last 8 bits could form → $2^8 = 256$ IP addresses
- AWS reserves 5 IPs → usable IPs = 251

ASSIGNING CIDR RANGES TO VPC AND SUBNETS

The **VPC** is assigned a CIDR block defining its entire IP address space.

VPC CIDR: 10.0.0.0/16 → range 10.0.0.0 – 10.0.255.255 (Total IPs: 65,536)

11111111 . 11111111 . 00000000 . 00000000



Subnets are smaller (non-overlapping) slices of the VPC’s CIDR block.

Subnet A: 10.0.1.0/24 → range 10.0.1.0 – 10.0.1.255 (Total IPs: 256)

Subnet B: 10.0.2.0/24 → range 10.0.2.0 – 10.0.2.255 (Total IPs: 256)

Subnet C: 10.0.3.0/24 → range 10.0.3.0 – 10.0.3.255 (Total IPs: 256)

11111111 . 11111111 . 11111111 . 00000000



Route Table

A Route Table is a set of rules that determine where outbound network traffic from a subnet is directed based on destination IP addresses.

Each route has two parts:

- **Destination** → Where the traffic wants to go (CIDR range)
- **Target** → Where AWS sends the traffic next (gateway or connection)

Destination	Target(next hop)
10.0.0.0/16	local
0.0.0.0/0	igw-123456

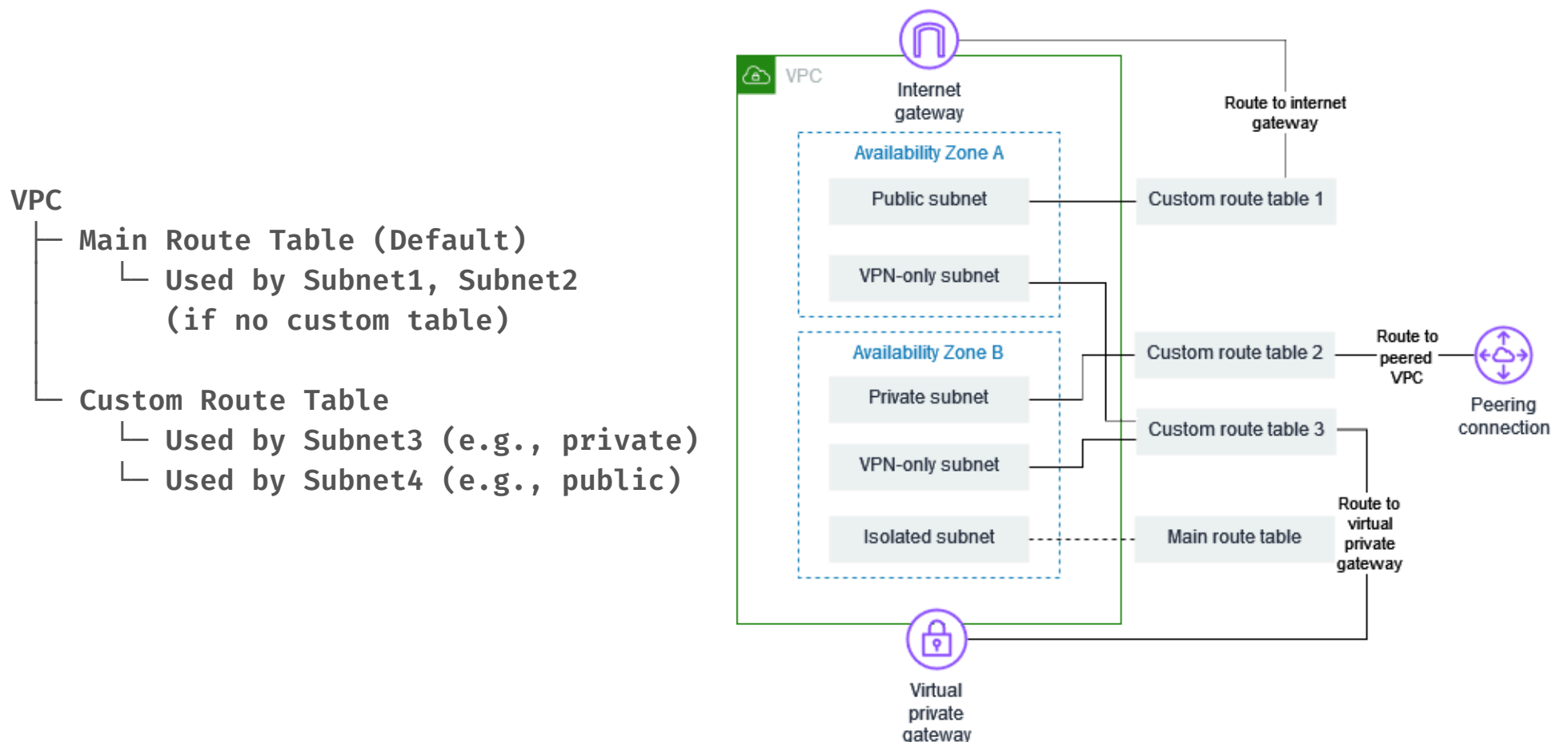
Any traffic within the VPC (10.0.0.0/16) stays local

Any traffic to the internet (0.0.0.0/0) is sent to the Internet Gateway

- Route tables belong to a VPC
- Route tables are associated with subnets
- One subnet uses one route table
- One route table can be used by multiple subnets

Default (Main) Route Table: Automatically created by AWS; used by subnets not explicitly associated with a custom table.

Custom Route Table: Manually created route table; can be associated with one or more subnets to control traffic independently.



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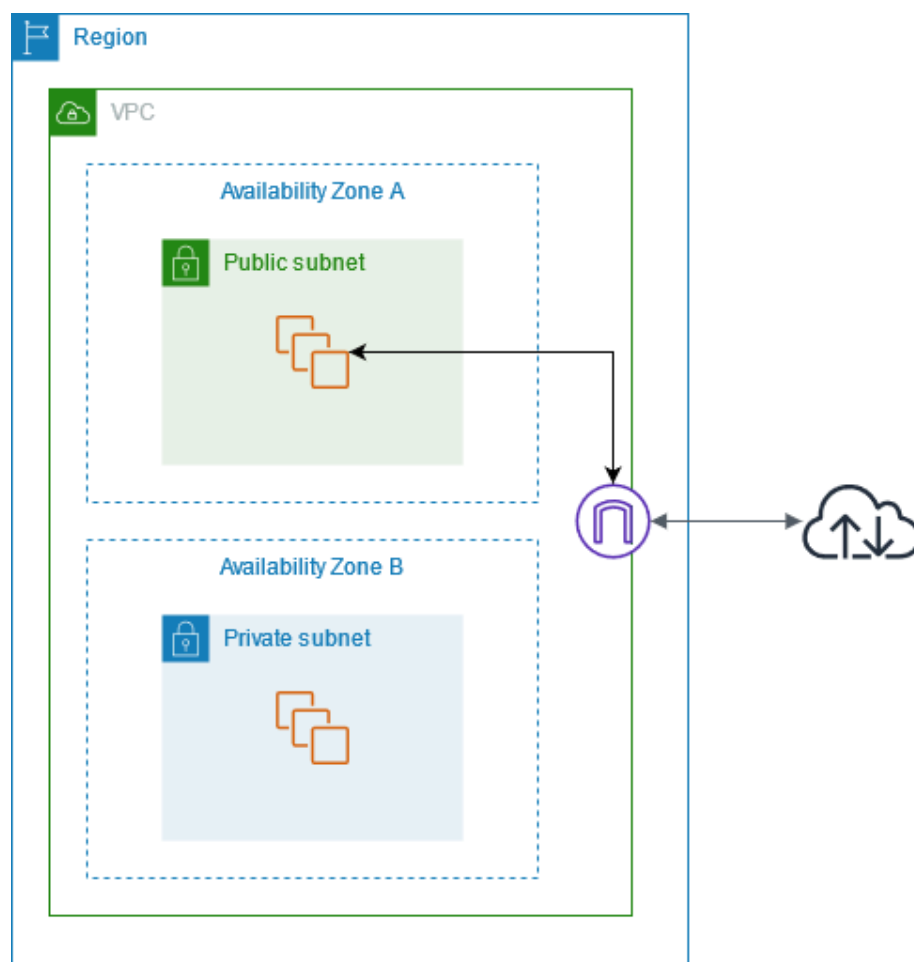
AWS Gateways

INTERNET GATEWAY (IGW)

A managed AWS gateway that allows communication between a VPC and the public internet.

- One IGW per VPC
- Enables inbound and outbound internet traffic (for instances with public IP)
- High availability & horizontally scaled by AWS
- Subnet becomes public when route table points 0.0.0.0/0 → IGW

Internet
↓
Internet Gateway
↓
Public Subnet → EC2 with
Public IP



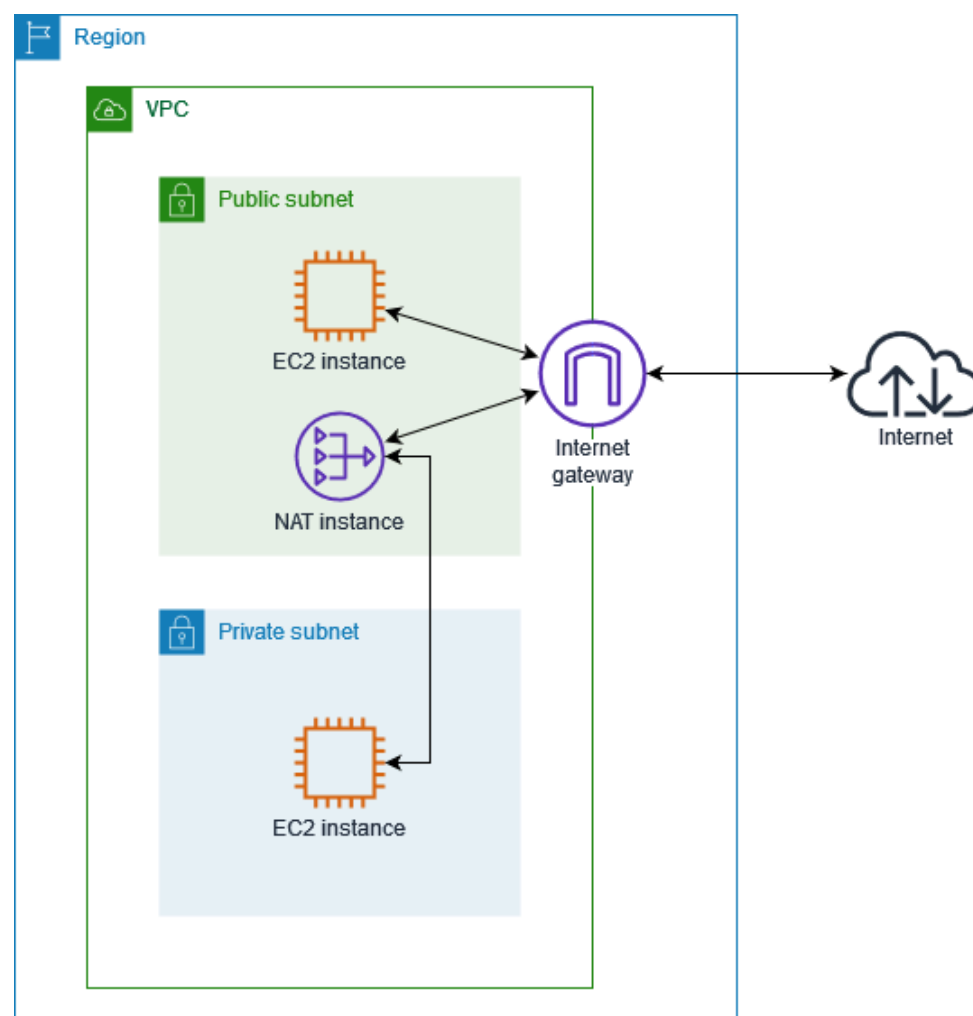
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NAT GATEWAY (OR NAT INSTANCE)

Allows instances in private subnets to access the internet for updates, downloads, or API calls without exposing them publicly.

- Placed in a public subnet
- Provides outbound internet access for private subnets
- Private instances remain unreachable from the internet
- Managed service → NAT Gateway is recommended over NAT Instance

Private Subnet → Route Table → NAT Gateway → IGW → Internet

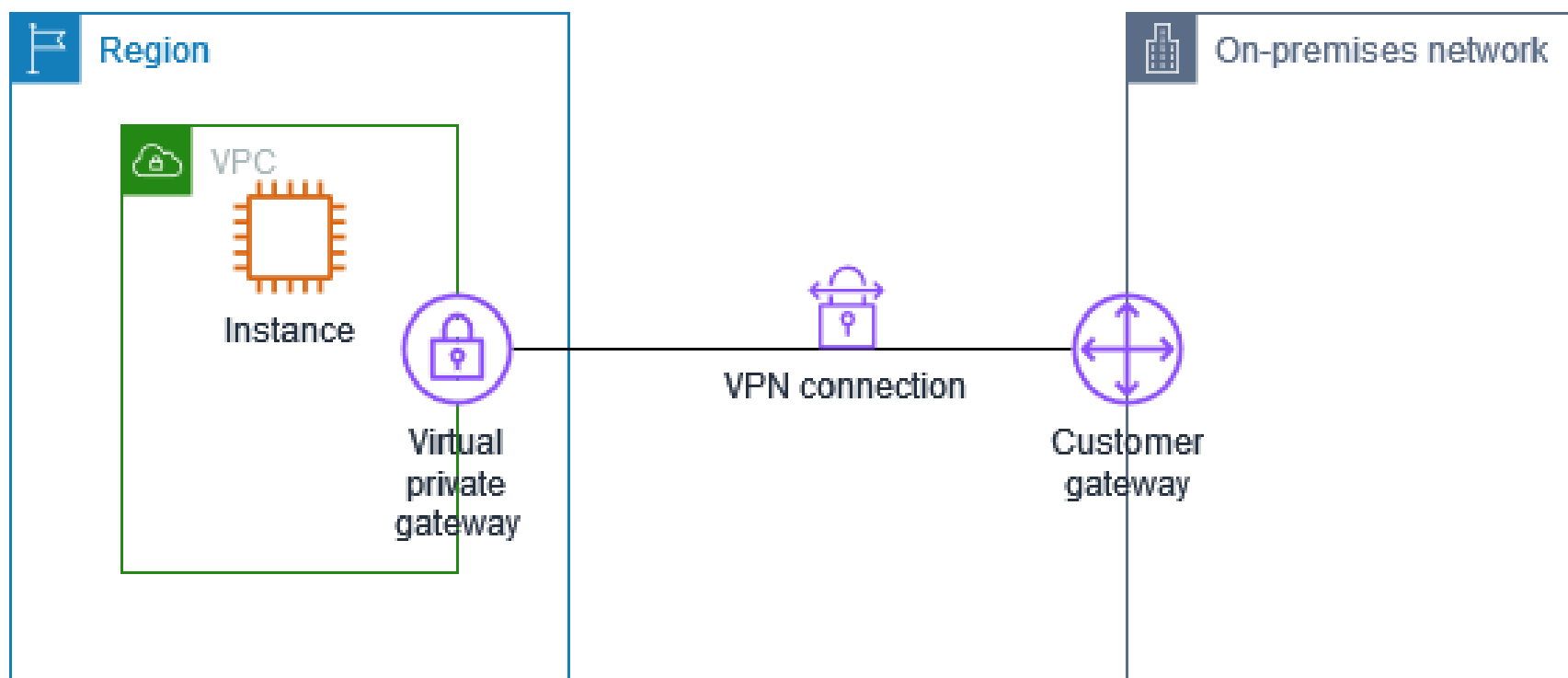
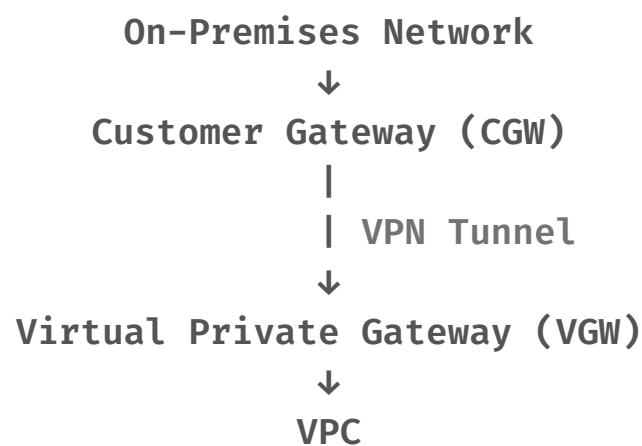


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VIRTUAL PRIVATE GATEWAY (VGW)

A VGW is a gateway on the AWS side that allows your VPC to connect securely to on-premises networks or remote networks via VPN.

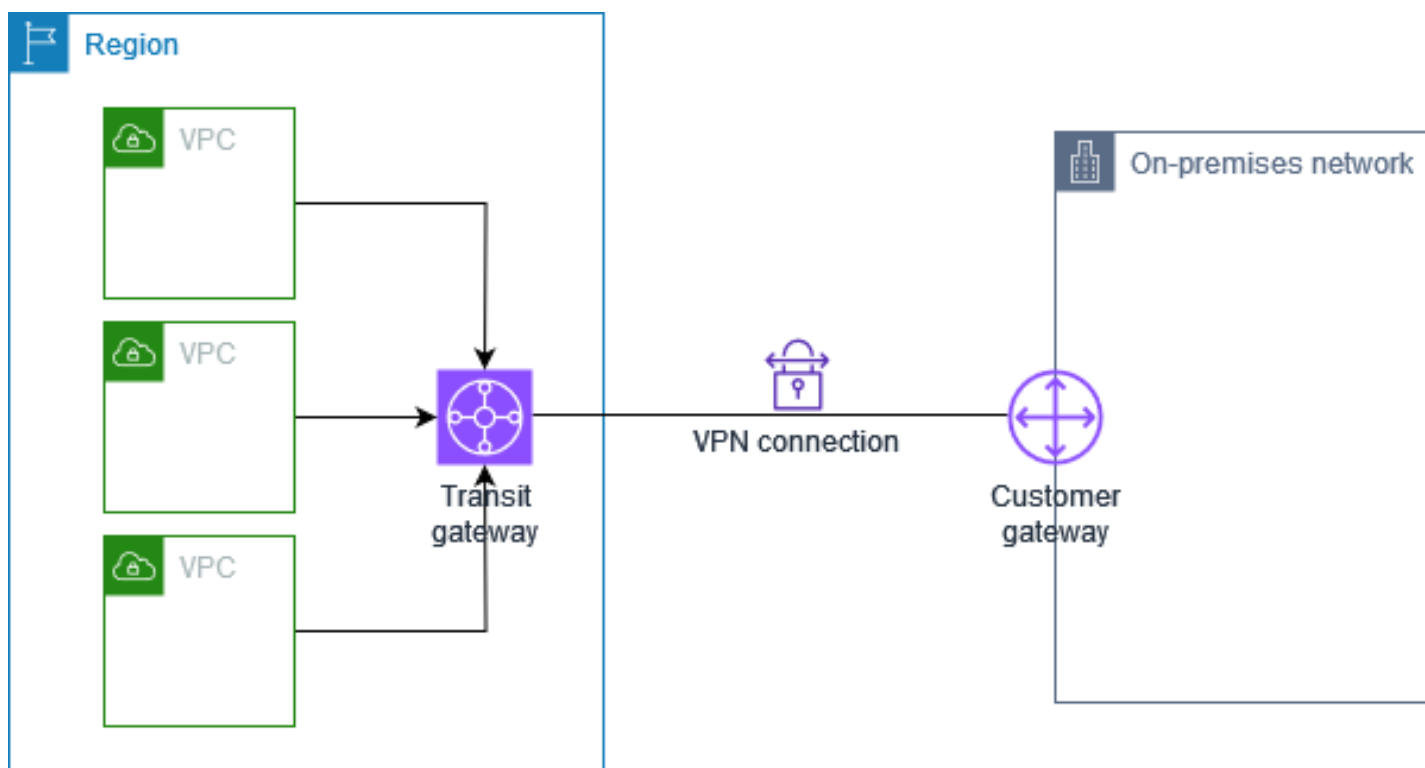
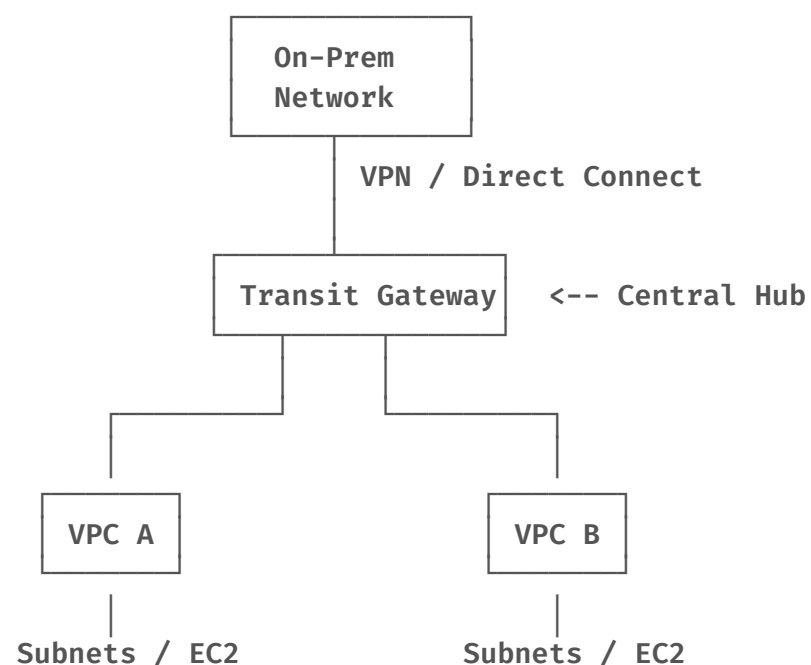
- Attached to a single VPC
- Works with Site-to-Site VPN connections
- Enables hybrid cloud architectures
- Provides secure, encrypted connectivity



TRANSIT GATEWAY (TGW)

A central hub that connects multiple VPCs and on-premises networks, replacing complex point-to-point VPC peering connections.

- Supports hundreds of VPCs and VPNs
- Simplifies network routing at scale
- Can connect VPCs in same or different regions
- Works with VGW, Direct Connect, and VPC attachments



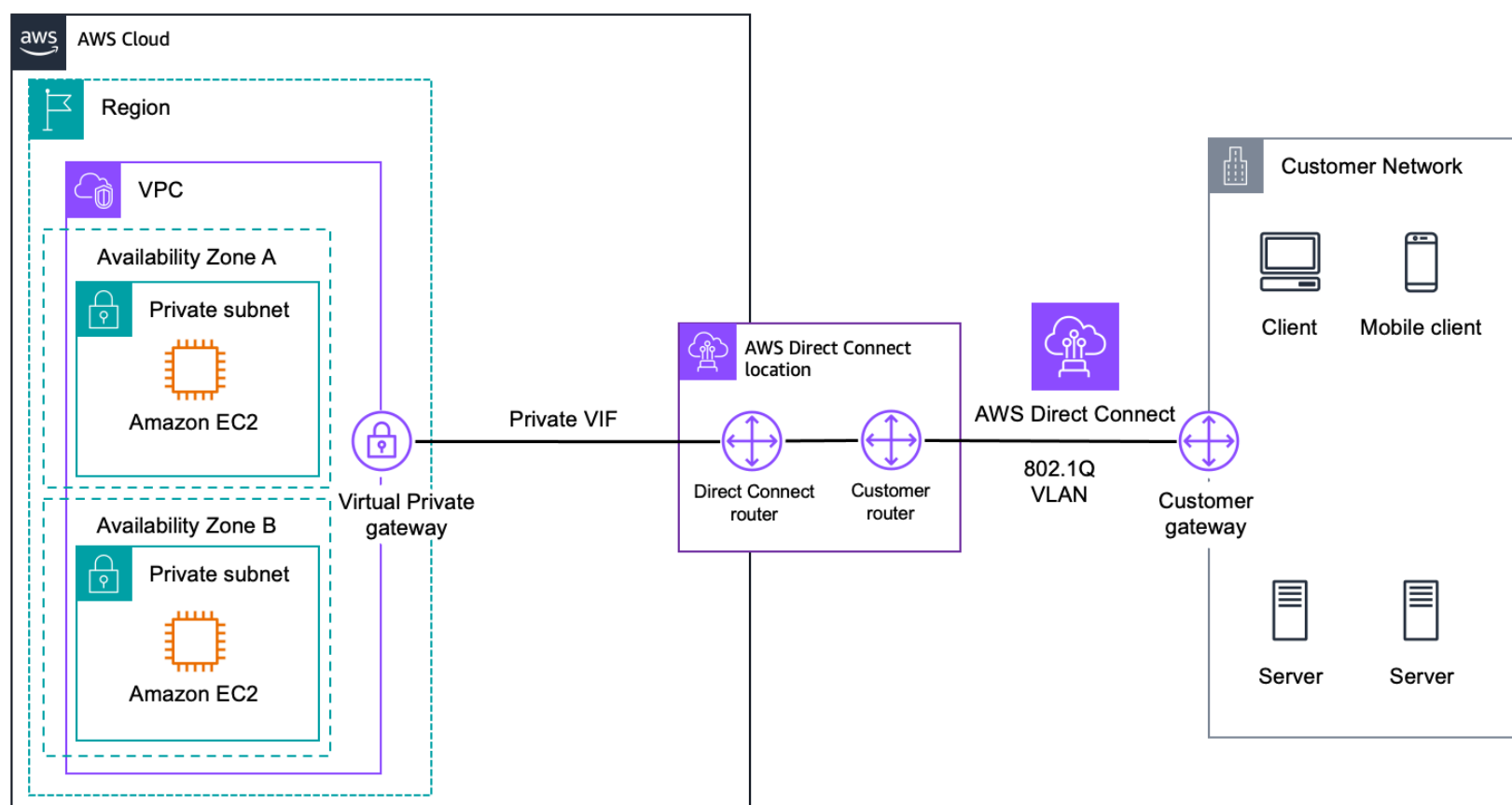
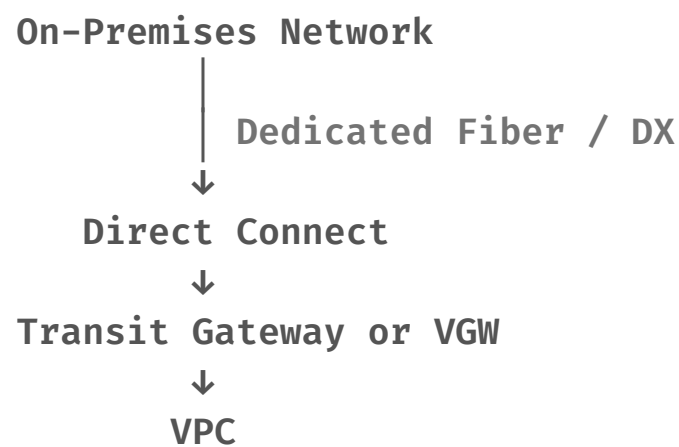
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Other Connections Types

DIRECT CONNECT (DX)

AWS Direct Connect is a dedicated private network connection from your on-premises data center to AWS.

- Provides low-latency, high-bandwidth connectivity
- Bypasses the public internet → more secure and stable
- Can be used with VGW or Transit Gateway
- Useful for hybrid cloud setups or large data transfers

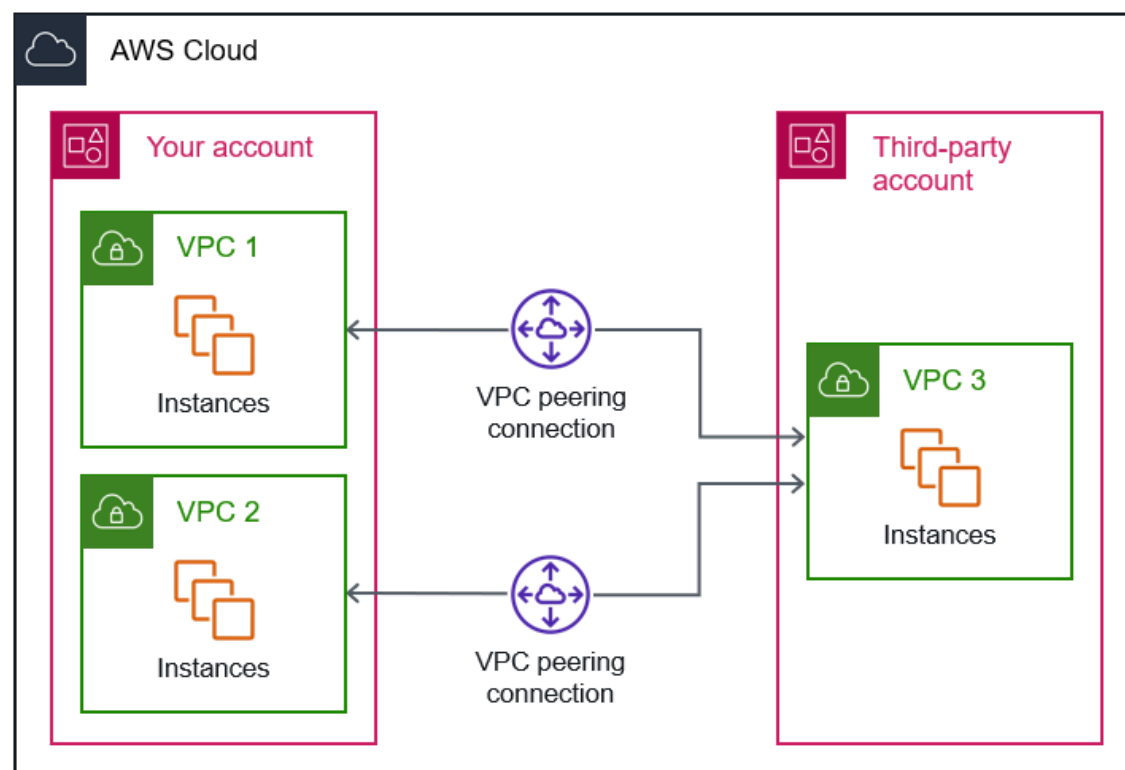
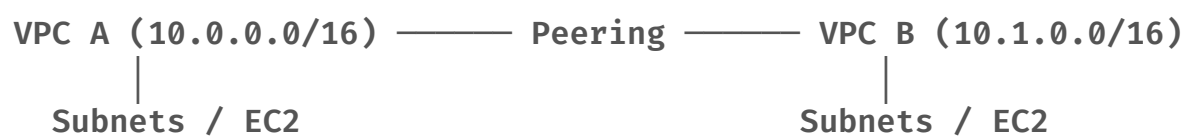


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VPC PEERING

A private network connection between two VPCs that enables them to communicate as if they are in the same network.

- VPCs can be in the same or different AWS regions
- Uses private IP addresses for communication
- Non-transitive: VPC A → B, B → C does not mean A → C automatically
- Route tables must be updated in both VPCs for traffic



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Security Groups vs NACLs

SECURITY GROUPS (SGS)

A virtual firewall at the resource level controlling inbound and outbound traffic.

- A security group can be attached to multiple instances.
- Stateful → if inbound is allowed, the response is automatically allowed outbound
- Controls ports and protocols
- Default SG allows all outbound, no inbound

A security group rule has 4 main components:

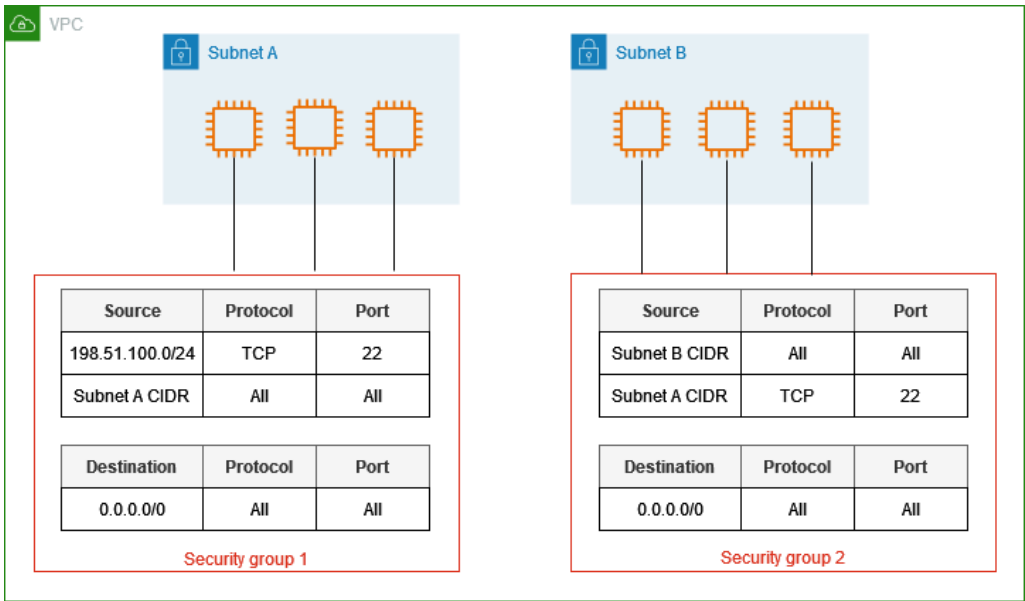
Component	Description
Type / Protocol	The protocol allowed, e.g., TCP, UDP, ICMP, or ALL
Port Range	Ports that are allowed, e.g., 80 for HTTP, 443 for HTTPS
Source / Destination	The IPs, CIDR, or security group that is allowed
Direction	Inbound (incoming) or Outbound (outgoing)

Example: Web Server SG

Direction	Protocol	Port	Source / Destination	Purpose
Inbound	TCP	80	0.0.0.0/0	HTTP traffic from internet
Inbound	TCP	443	0.0.0.0/0	HTTPS traffic from internet
Outbound	ALL	ALL	0.0.0.0/0	Allow all outbound traffic

Inbound
Internet → [Security Group] → EC2 Instance

Outbound
EC2 Instance → [Security Group] → Internet or other resources



NETWORK ACLS (NACLs) — AWS SUBNET-LEVEL FIREWALL

A stateless firewall at the subnet level controlling inbound and outbound traffic.

- Associated with subnets, not individual instances
- Stateless: return traffic must have its own explicit rule.
- Evaluated in numerical order of rules (lowest rule number first)
- Default NACL allows all traffic in and out
- Can block unwanted traffic at subnet level for extra security

A **NACL rule** has 6 main components:

Component	Description
Rule Number	Determines the evaluation order (lowest number first).
Protocol	The network protocol (TCP, UDP, ICMP, or ALL).
Port Range	The specific ports affected by the rule (e.g., 22 for SSH, 80 for HTTP).
Source / Destination	The IP range (CIDR block) the rule applies to. For inbound, it's source ; for outbound, it's destination .
Allow / Deny	Whether traffic matching this rule is allowed or blocked .
Direction	Indicates whether the rule applies to inbound or outbound traffic.

Example: NACL Rule Structure

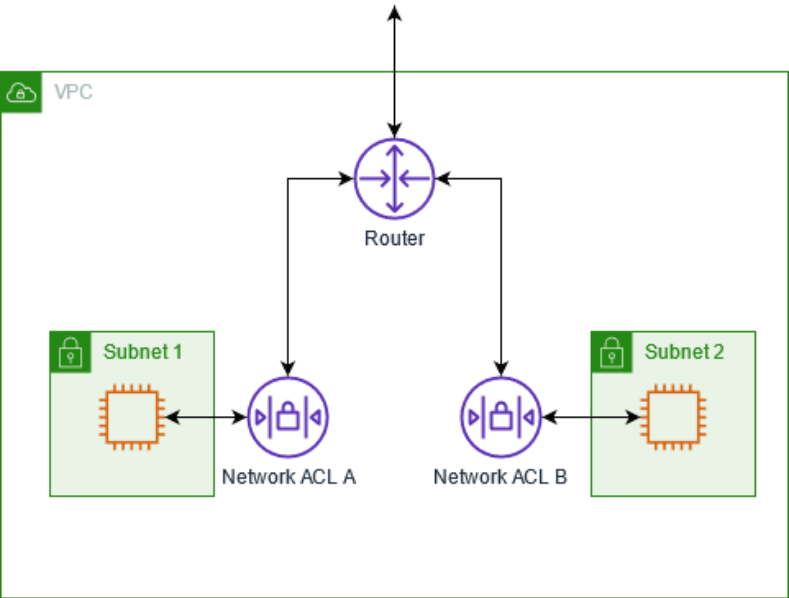
Direction	Rule #	Protocol	Port Range	Source / Destination	Allow/Deny
Inbound	100	TCP	80	0.0.0.0/0	Allow
Inbound	110	TCP	22	203.0.113.0/24	Allow
Inbound	*	ALL	ALL	0.0.0.0/0	Deny
Outbound	100	ALL	ALL	0.0.0.0/0	Allow

Inbound Traffic

Internet —> NACL (subnet-level firewall) —> Subnet —> EC2 Instances

Outbound Traffic

EC2 Instances —> NACL (subnet-level firewall) —> Internet



THANK YOU !
FOR YOUR TIME

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